

D10 SW1-SW20: front-panel order and matrix wiring

BENCH DEVELOPMENT ONLY - NOT ASSEMBLY APPROVED - SCHEMATIC REVIEW REQUIRED

Front/button view: one physical line, SW1 left to SW20 right. Cells show SW, label, CAN ID.

SW1	SW2	SW3	SW4	SW5	SW6	SW7	SW8	SW9	SW10	SW11	SW12	SW13	SW14	SW15	SW16	SW17	SW18	SW19	SW20
LIDAR	AUTOPILOT	EMERG COM	CM5 BOOT	MAP	COM	NAV	ENGINE	TRAFFIC	PFD	DIRECT TO	FLIGHT PLAN	AIRPORT	COM SWAP	NAV SWAP	XPDR	BARO	ALERT ACK	BACK	MENU / DIM
0x601	0x602	0x603	0x604	0x605	0x606	0x607	0x608	0x609	0x60A	0x60B	0x60C	0x60D	0x60E	0x60F	0x610	0x611	0x612	0x613	0x614

Electrical matrix (not physical rows): ID = SW - 1; row = ID / 4; column = ID % 4.

ROW0 J1-9 -> GPIO0 SW1 SW2 SW3 SW4

ROW1 J1-10 -> GPIO1 SW5 SW6 SW7 SW8

ROW2 J1-12 -> GPIO2 SW9 SW10 SW11 SW12

ROW3 J1-2 -> GPIO3 SW13 SW14 SW15 SW16

ROW4 J1-4 -> GPIO4 SW17 SW18 SW19 SW20

COL0 J1-1 -> GPIO5; COL1 J1-3 -> GPIO6; COL2 J1-5 -> GPIO7; COL3 J1-7 -> GPIO8

J1-6 -> common GND -> all LED pad 5; J1-8 -> protected switched +5 V -> LED pad 6

Switch pads 3/4 -> row; pads 1/2 -> diode -> column. Verify active-low diode polarity.

CAN1 TX22/RX23 -> 3.3 V-compatible transceiver -> CANH/CANL; 500 kbit/s; bus-end termination.

Rear view: J1 on right; pin 1 lower-right, pin 2 above; numbering increases leftward.

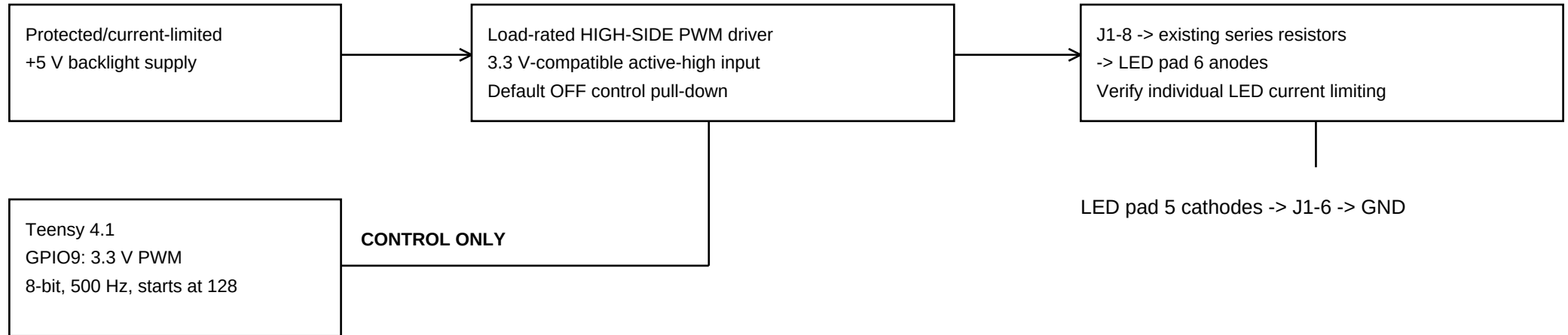
14: NO PAD	12: ROW2	10: ROW1	8: +5V	6: GND	4: ROW4	2: ROW3
13: NO PAD	11: NO PAD	9: ROW0	7: COL3	5: COL2	3: COL1	1: COL0

User-supplied R3 mapping only: verify every net, missing-pad alignment and LED polarity unpowered.

Safe backlight power path (conceptual)

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Teensy GPIO is 3.3 V only. GPIO9 must NEVER connect directly to J1-8 (+5 V).



Common GND: Teensy, driver and J1-6. Keep PWM control disconnected until driver review.

Do NOT put an N-MOS drain on LED anodes and source on GND: it shunts the supply.

Do NOT connect a +5 V gate pull-up to GPIO9; a high-side P-MOS needs a level-shifting stage.

A level shifter or optocoupler alone is not a load-rated power switch or LED current regulator.

A blanket 100 ohm resistor is not verified protection for 20 parallel LEDs.

Verify current/inrush, component ratings, fuse/current limiting, duty polarity and PWM suitability.

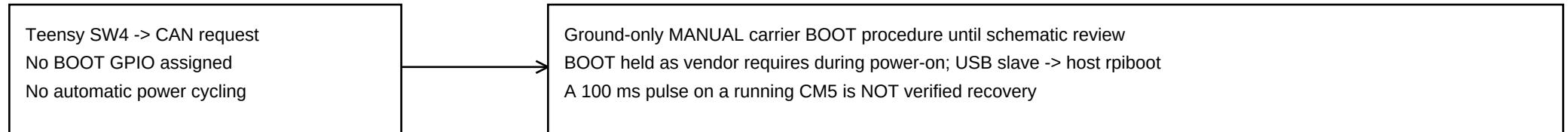
Bench-test duty 0/128/255; verify GPIO never exceeds 3.3 V and driver defaults OFF at reset.

No carrier/PCB schematic inspected. Do not assemble or use in flight from this concept diagram.

CM5 maintenance and independent COM integration boundaries

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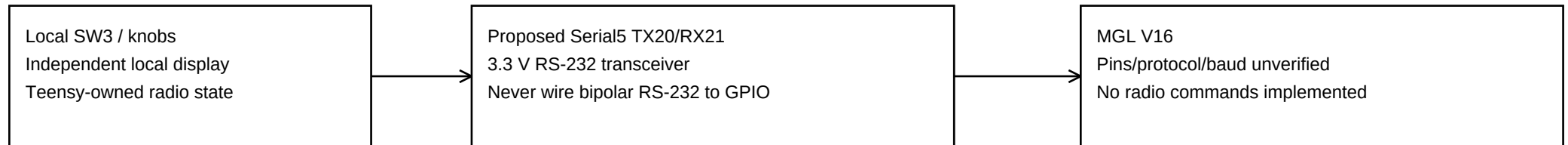
SW4 / 0x604: maintenance request only - not soft reboot, reset, power switch or flash command.



Future only: verified isolated relay contacts or open-drain transistor across BOOT/GND.

Review carrier revision, pull-up voltage, power sequencing, isolation and physical ground interlock.

SW3 / 0x603: request only in current firmware - a CAN request is NOT an independent radio.



CM5 is only an optional status/logging observer during autonomous control; it is not the control path.

Required: verify V16 ICD, local display and knobs, EEPROM CRC/version, radio ACKs and ownership handoff.

Acceptance: remove CM5 power; verify local tuning/display, corrupted storage handling and safe resume.

These interfaces remain UNIMPLEMENTED / UNVERIFIED pending detailed design and schematic review.

Protocol, full mapping, references and bench checklist: [button-matrix/TEENSY_FIRMWARE.md](#)